
IAL Physics Unit 1 (WPH11): A* Revision Guide

Mechanics and Materials

Compiled from 22 WPH11 papers and their mark schemes (January 2019 – June 2026). Mark-scheme wording is quoted where the exact phrase earns the mark.

1. The paper and where the marks are

Format. 1 hour 30 minutes, 80 marks. Section A: 10 multiple-choice questions (1 mark each). Section B: about 9–10 structured questions (70 marks), always including one asterisked 6-mark extended-response question. Almost every Section B question is set in a real context (bridge, skydiver, crane, submarine) and many end with “Is the claim correct?”.

Topic	How often (22 papers)	What is actually asked
Projectiles	Almost every paper	Resolve the launch velocity, find time of flight from one direction, use it in the other, then decide “does it clear the wall / reach the target?”.
Upthrust, Stokes’ law, terminal velocity	Every paper	Free-body diagram (W , U , D); $W = U + D$; $F = 6\pi\eta r v$; conditions for Stokes’ law; effect of temperature on viscosity; floating.
Stress, strain, Young modulus	Every paper	E from a gradient or data; extension from E ; breaking force; area under graphs; method for measuring E .
Moments and equilibrium	Every paper	Principle of moments with angled forces; hinge/support forces; toppling; centre of gravity definition.
Momentum	Almost every paper	Principle of conservation (2-mark statement); collisions/explosions; KE lost; link to Newton’s 3rd law.
Work, energy, power, efficiency	Every paper	ΔE_{grav} , E_{k} , work against resistive force, $P = W/t = Fv$, efficiency, and an “assess the claim” conclusion.
Newton’s laws and forces	Every paper	N3 pairs (MCQ every year); $\Sigma F = ma$ with weight; lifts and trampolines; slopes; vector triangles (scale drawing).
Kinematics graphs and suvat	Every paper	Gradient and area of s - t / v - t graphs; tangents; measuring g ; 6-mark graph description.
Springs, Hooke’s law, elastic energy	Most papers	k from a graph; springs in series/parallel; $E_{\text{el}} = \frac{1}{2}F\Delta x$; bungee and catapult energy.

2. How A* scripts collect marks

RULES TAKEN FROM THE MARK SCHEMES

1. **“Show that”**: give the final answer to **one more significant figure** than the value printed. A reverse calculation earns at most 2 marks (none of the final mark on a 3-mark question). If you cannot get the value, **use the printed value** next.
2. **Units**: a missing or wrong unit loses the final mark. **$g = 9.81 \text{ N kg}^{-1}$** (9.8 accepted); 10 loses a mark.
3. **“Is it correct?” / “Will it...?”** The last mark needs your number, a **comparison** with the value in the question (“ $2.1 \text{ m} < 3.0 \text{ m}$ ”) and a **conclusion** (“so the rider clears the river”). A correct conclusion from your own wrong value still scores (ecf).
4. **Scale drawings**: when a question says “draw a scale diagram”, a trigonometry answer earns only the final mark. Draw arrows at least 6–8 cm long with a stated scale, label at least two sides, and point arrows nose-to-tail.
5. **Free-body diagrams**: –1 mark for each extra force; every arrow must **touch the dot**; label with names (weight, upthrust, drag, normal contact force); make lengths match the situation (balanced forces drawn equal).
6. **Graphs**: use a **large triangle** (over half the drawn line) for gradients; draw a **tangent** for instantaneous rates; count squares for areas and state the value of one square.
7. **Words that are refused**: “net force” is accepted but “resultant force” is safer; “equal and opposite reaction” is *not* accepted for N3 pairs; “human error” is ignored (say “reaction time → random error”); “calculate” the intensity/time is refused where “measure” is meant.

6-MARK EXTENDED RESPONSE: HOW IT IS SCORED

Six indicative content (IC) points. Marks for content: 6 IC → 4, 5–4 IC → 3, 3–2 IC → 2, 1 IC → 1, plus up to 2 marks for linkage. **6/6 needs all six points as a connected chain.**

- Write 6–7 short numbered sentences joined with “so”, “therefore”, “because”. Name the law you are using (“By Newton’s third law...”).
- Most Unit 1 six-markers are one of five types: forces → acceleration → terminal velocity; Newton’s 3rd + 2nd law in an interaction; energy transfers in stages; a practical method; reading a graph. Model answers for all of them are in section 10.

3. Quantities, units and vectors

- **Vectors**: displacement, velocity, acceleration, force (including weight, upthrust, drag), momentum, moment. **Scalars**: distance, speed, mass, time, energy, work, power, density, pressure, stress, strain, Young modulus, viscosity, efficiency, temperature. This is an MCQ in nearly every paper.
- **Base units**: $\text{N} = \text{kg m s}^{-2}$; $\text{J} = \text{kg m}^2 \text{s}^{-2}$; $\text{W} = \text{J s}^{-1} = \text{kg m}^2 \text{s}^{-3}$; $\text{Pa} = \text{N m}^{-2} = \text{kg m}^{-1} \text{s}^{-2}$; momentum $\text{kg m s}^{-1} = \text{N s}$; viscosity Pa s ; $k \text{ N m}^{-1}$; $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$. Distinguish them: kg m s^{-1} is momentum, kg m s^{-2} is force, $\text{kg m}^2 \text{s}^{-2}$ is energy.

- **Prefixes:** $\mu = 10^{-6}$, $m = 10^{-3}$, $c = 10^{-2}$, $k = 10^3$, $M = 10^6$, $G = 10^9$. Convert mm and cm to m **before** calculating stress, area or volume. Square or cube the conversion for area ($1 \text{ mm}^2 = 10^{-6} \text{ m}^2$) and volume.
- **Resolving:** horizontal = $F \cos \theta$, vertical = $F \sin \theta$ (with θ to the horizontal). On a slope the component of weight **along** the slope is $mg \sin \theta$; perpendicular is $mg \cos \theta$.
- **Adding vectors:** at right angles, use Pythagoras and tan for the angle. Otherwise use a scale drawing (triangle or parallelogram). **Resultant force:** “the (vector) sum of all the forces acting on an object” or “the single force with the same effect as all the forces”.
- **Displacement vs distance:** displacement is the straight line from start to end (vector); distance is the length of the path (scalar).

4. Kinematics

$v = u + at$; $s = \frac{1}{2}(u + v)t$	suvat is valid only for constant acceleration . On a curved slide the gradient (and so a) varies, so use energy instead (Jan 2024).
$s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$	Choose a sign convention (up = +, so $a = -9.81$). With a quadratic in t , use the formula and choose the sensible root (Jun 2022).
s - t graph	Gradient = velocity (tangent for instantaneous). Straight line = constant velocity; parabola = constant acceleration; average velocity = total displacement $\div$ total time.
v - t graph	Gradient = acceleration; area = displacement (area below the axis is negative displacement). Area under an a - t graph = change in velocity.

Measuring g (practical, regular 4-6 marks)

- **Light gate:** an object (card or rod) of measured length passes through the light gate; $v = \text{length} \div \text{time it blocks the beam}$. Measure the drop height s to the gate; repeat at several heights and average. Plot v^2 against s : gradient = $2g$ (or v^2 against $2s$: gradient = g).
- **Timing a fall from rest:** $s = \frac{1}{2}gt^2$, so plot s against t^2 : straight line through the origin, **gradient = $g/2$** . A stopwatch involves human reaction time, which causes large random errors.
- **Strobe photograph:** a laboratory strobe with a shorter interval gives smaller movement per flash, so the uncertainty is lower.
- Air resistance negligible? Calculate a and compare it with g (Jun 2025: $9.78 \text{ m s}^{-2} = 0.997 g > 0.95 g$, so yes).

Projectiles: the routine (4-6 marks, almost every paper)

1. Resolve: $u_H = u \cos \theta$, $u_V = u \sin \theta$.
2. Horizontal: $a = 0$, so $s = u_H t$. Vertical: $a = -9.81 \text{ m s}^{-2}$.
3. **Time links the two directions.** Find t from whichever direction has enough data, then use it in the other.
4. Maximum height: $v_V = 0$, so $s = u_V^2/2g$. Add the launch height if asked for height above the ground.
5. Compare with the target (wall height, distance to boat, ring position) and conclude.

EXAMPLE (OCT 2022). Ball at 9.70 m s^{-1} , 49° to the horizontal, wall 5.00 m away and 3.00 m high. $u_H = 6.36 \text{ m s}^{-1}$, so $t = 5.00/6.36 = 0.79 \text{ s}$; $u_V = 7.32 \text{ m s}^{-1}$, so $s = 7.32(0.79) - \frac{1}{2}(9.81)(0.79)^2 = 2.72 \text{ m} < 3.00 \text{ m}$, so the ball does not clear the wall.

- **Horizontal launch:** $u_V = 0$, so the time of fall depends only on height. Balls launched horizontally at different speeds land at the **same time**.
- **Faster horizontal launch:** the horizontal velocity is larger but the vertical landing velocity is unchanged, so the landing angle to the horizontal is smaller.
- **Landing velocity:** combine v_H and v_V by Pythagoras; angle = $\tan^{-1}(v_V/v_H)$.
- **Effect of air resistance** (2-3 marks, also a 6-marker): drag opposes the motion. There is a horizontal deceleration, so the horizontal velocity decreases. The vertical deceleration on the way up is greater, so the maximum height is lower and the time in the air is shorter. So the range is shorter, and the path is no longer a symmetrical parabola.
- **Why range has a maximum:** a larger angle gives more time in the air but a smaller horizontal velocity; range = time $\times$ horizontal velocity.

5. Forces, Newton's laws and momentum

Newton's 1st law	An object stays at rest or moves with constant velocity unless a resultant force acts on it.
Newton's 2nd law	$\Sigma F = ma$: acceleration is proportional to the resultant force (for constant mass). Always find the resultant first.
Newton's 3rd law pair	If A exerts a force on B, B exerts a force on A that is equal in magnitude, opposite in direction , of the same type , and acts on a different body (at the same time). Weight and normal contact force on the same object are not a pair: they act on the same object and are different types. The pair of the Earth's pull on you is <i>your</i> gravitational pull on the Earth.

- **Lift/trampoline/rocket:** write $\Sigma F = (\text{upward force}) - mg = ma$. Scales read the normal contact force: more than mg when accelerating up, less when accelerating down (or decelerating while going up). Rocket: $a = (\text{thrust} - mg)/m$, and a increases as fuel is used because the mass decreases.
- **Connected masses (MCQ Oct 2025):** $a = (\text{resultant force})/(\text{total mass being accelerated})$, e.g. $(0.5 - 0.4)g/(0.5 + 0.4)$.

- **Slope:** frictionless $a = g \sin \theta$, independent of mass. Constant speed means the component of weight down the slope = total resistive force.
- **Equal decelerations (Jan 2025):** if the resistive force $\propto$ mass, then $a = F/m$ is the same for both vehicles.

Momentum

PRINCIPLE OF CONSERVATION OF MOMENTUM (2 MARKS, ABOUT HALF OF ALL PAPERS)

"The total momentum before a collision equals the total momentum after (the total momentum of a system remains constant) (1), **provided no external (resultant) force acts** / in a closed system (1)." The second mark is the one most often lost.

- **Collisions:** pick a positive direction, write the momentum of each object with its sign, and set total before = total after. A negative answer means the object moves the opposite way; say so.
- **Explosions / recoil (skateboard, ball machine, rocket gases):** the initial momentum is zero, so the two parts have equal and opposite momenta. The heavier part moves more slowly.
- **Kinetic energy is usually not conserved.** Find the KE lost as total $\frac{1}{2}mv^2$ before – after. Explain it as: work is done by forces during the collision (deformation), transferring energy to thermal energy, and this is not recovered.
- **Force from momentum:** $F = \Delta p / \Delta t$. Rate of change of momentum of a jet = (mass per second) $\times$ velocity (Jan 2021).
- **Momentum checks (Oct 2022):** calculate the total before and after; if they differ, an external force (friction) acted.
- **Link to N3 (frequent):** A exerts a force on B and B exerts an equal and opposite force on A for the same time. So the momentum changes are equal and opposite and total momentum is conserved; the lighter object has the larger velocity change.

EXAMPLE (OCT 2021). 2.7 kg at 10 m s⁻¹ hits 7.9 kg at rest; 7.9 kg moves off at 5.0 m s⁻¹.
 $27.0 = 2.7v + 39.5$, so $v = -4.6$ m s⁻¹ (the 2.7 kg mass rebounds).

6. Moments, centre of gravity and equilibrium

Moment	Force $\times$ perpendicular distance from the pivot to the line of action of the force. With an angled force use $F \sin \theta \times d$ (or $F \times d \sin \theta$).
Principle of moments	For an object in equilibrium, the sum of clockwise moments about any point equals the sum of anticlockwise moments about that point.
Equilibrium (2 marks)	Zero resultant force (in any direction) and zero resultant moment (about any point).
Centre of gravity	"The point at which the weight of the object may be taken (considered) to act." A uniform beam has its centre of gravity at its midpoint.

Toppling	The object topples once the line of action of its weight passes outside its base (beyond the pivot), because the weight then gives an unbalanced moment in the same direction as the disturbing force.
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ROUTINE: TWO UNKNOWN SUPPORT FORCES (BEAM, BRIDGE, SHELF, VEHICLE)

1. Take moments about one support, so that its unknown force has zero moment, and find the other support force.
2. Use $\Sigma F = 0$ (total up = total down) to find the first, or take moments about the second support.
3. Hinge forces: find the horizontal and vertical components separately, then $F = \sqrt{F_H^2 + F_V^2}$ and the angle by tan.

EXAMPLE (JUN 2024). Bridge 120 m long, lorry 4.2×10^5 N at 35 m, bridge 9.8×10^5 N at 60 m. Moments about P: $Q \times 120 = 4.2 \times 10^5 \times 35 + 9.8 \times 10^5 \times 60$, so $Q = 6.1 \times 10^5$ N; then $P = 4.2 \times 10^5 + 9.8 \times 10^5 - 6.1 \times 10^5 = 7.9 \times 10^5$ N.

- **“How do the forces change?” (3-4 marks):** as the load moves away from X its moment about X increases; the sum of moments stays zero, so the moment of the other support must increase. Its distance is fixed, so that force increases (and the total stays equal to the total weight).
- **Angled push or pull:** the same moment needs a larger force when the force is applied at a smaller angle to the lever, because the perpendicular distance (or component) is smaller.
- **Crane/counterweight:** moving the load out increases the clockwise moment, so the counterweight must move further out. Check by solving for the maximum load distance.
- **Boat stability:** the moment of the weight about the centre of buoyancy restores the boat upright; if the weight’s line of action moves past the upthrust’s, the moment reverses and the boat capsizes.

7. Work, energy, power and efficiency

$\Delta W = F\Delta s$ (Δs in the direction of F)	With an angle, $\Delta W = F\Delta s \cos \theta$. Increasing the angle reduces the horizontal component, so less work is done per metre.
$E_k = \frac{1}{2}mv^2$, $\Delta E_{\text{grav}} = mg\Delta h$	Δh is the vertical height: on a slope, $\Delta h = \text{length} \times \sin \theta$. Doubling E_k multiplies v by $\sqrt{2}$.
$P = \Delta W/\Delta t = Fv$	Driving force at constant power: $F = P/v$, so the force decreases as speed increases.
efficiency = useful output $\div$ total input	Energy or power, never $\times 100$ unless giving a percentage. Total input = useful + wasted.

ROUTINE: ENERGY WITH A RESISTIVE FORCE (SLOPE, RAMP, HAMMER, CYCLIST)

Write the energy statement first, then substitute: **energy at start = energy at end + work done against resistive force**. For example, $\frac{1}{2}mv^2 = mg(L \sin \theta) + F_R L$ (Jan 2019: $L = 120$ m). Work done by the person or motor adds to the input.

- **Speed at the bottom is independent of mass** (frictionless): $mg\Delta h = \frac{1}{2}mv^2$, so $v = \sqrt{2g\Delta h}$, assuming the work done against resistive forces is negligible.
- **Counterweights** (lift, bridge): the counterweight's loss of GPE supplies part of the energy, so the motor does less work (and needs only to overcome resistive forces).
- **Regenerative braking and drag**: at higher speeds drag is larger, so more work is done against air resistance, so a smaller fraction of KE is recovered and the efficiency is lower.
- **Pumping water**: power = $(mg\Delta h + \frac{1}{2}mv^2)$ per second. Mass per second = density $\times$ volume per second = ρAv .

8. Fluids: density, upthrust, viscosity

$\rho = m/V$; sphere $V = \frac{4}{3}\pi r^3$; cylinder $V = \pi r^2 l$	Use the radius , not the diameter (MCQ trap). Halve the diameter first.
Upthrust = weight of fluid displaced = $\rho_{\text{fluid}} V_{\text{submerged}} g$	Upthrust depends on the fluid density and the submerged volume, not on the object's density.
Stokes' law $F = 6\pi\eta r v$	Valid for a small, spherical object moving at low speed with laminar flow . All four conditions appear in MCQs; low viscosity and a cylindrical shape are not conditions.
Terminal velocity: $W = U + D$	Falling sphere: $D = W - U$. Rising bubble or balloon: $U = W + D$. Some questions use $D = kv^2$ for large, fast objects.

Laminar flow	Layers of fluid flow past each other without mixing; the velocity at any fixed point stays constant. Turbulent : the layers mix and eddies form, with the velocity at a point changing.
Viscosity and temperature	For liquids, a higher temperature means a lower viscosity, so less drag at a given speed and a higher terminal velocity (less time to fall). Cold oil gives the reverse.
Terminal velocity	"The constant (maximum) velocity reached when the resultant force is zero (drag + upthrust = weight)."
Floating	The upthrust equals the weight of the object (resultant force zero), so the weight of fluid displaced equals the object's weight. In denser water (seawater) a smaller volume needs to be displaced, so it floats higher.

Sinking or rising	Compare U with W : if $U > W$ there is a resultant upward force and it accelerates up. Pumping water into a submarine increases W (with U constant), so it sinks.
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ROUTINE: STOKES' LAW / VISCOSITY CALCULATION (4-6 MARKS, EVERY PAPER)

- $V = \frac{4}{3}\pi r^3 \rightarrow W = \rho_{\text{sphere}} Vg$ and $U = \rho_{\text{fluid}} Vg$.
- At terminal velocity, $D = W - U$.
- $\eta = D/(6\pi r v)$, or $v = D/(6\pi \eta r)$. Terminal v = distance $\div$ time between markers.
- "Does Stokes' law apply?" Calculate $6\pi \eta r v$ and compare with $W - U$ (or compare speeds). A large mismatch means the flow is not laminar or the speed is not low.

EXAMPLE (JUN 2021). $r = 0.0175$ m, $\rho_s = 2520$, $\rho_f = 1430$ kg m⁻³; $V = 2.24 \times 10^{-5}$ m³; $W = 0.554$ N, $U = 0.314$ N, so $D = 0.240$ N. Shortcut: $D = (\rho_s - \rho_f)Vg$.

Falling-ball viscometer (practical)

- Put rubber bands/markers on the tube, the first far enough below the surface for terminal velocity to have been reached (check by timing two equal sections: the times should be equal).
- Time the fall between the markers with a stopwatch; measure the distance with a metre rule; repeat and average. v = distance $\div$ mean time (or the gradient of distance against time).
- Measure the diameter with a micrometer (several places), the ball's mass on a balance, and the oil's density. Keep the temperature constant and use a wide tube.
- A larger ball falls faster, so the time is shorter and the reaction-time uncertainty becomes a larger fraction of it.
- Why does v_{terminal} increase with size? Weight and upthrust $\propto r^3$ but drag $\propto r$, so the forces balance only at a higher speed.
- Starting the stopwatch after the ball has passed the marker gives a shorter time, so a larger speed and a smaller calculated η .

9. Materials

Hooke's law $\Delta F = k\Delta x$	k = gradient of a force-extension graph (stiffness, N m ⁻¹). Holds up to the limit of proportionality .
$E_{\text{el}} = \frac{1}{2}F\Delta x = \frac{1}{2}k\Delta x^2 = F^2/2k$	Area under the force-extension graph. Between two loads, use the trapezium area, not $\frac{1}{2} \times \text{final force} \times \text{change in extension}$.
$\sigma = F/A$, $\epsilon = \Delta x/x$, $E = \sigma/\epsilon$	$A = \pi d^2/4$. $E = Fx/(A\Delta x)$ = gradient $\times x/A$ (Young modulus is a material property).
$k = EA/x$	A longer wire stretches more (smaller k); a thicker wire stretches less (larger k). Same material and stress means same strain.

Area under σ - ϵ graph = energy per unit volume	$\frac{1}{2}\sigma\epsilon = \frac{1}{2}(F/A)(\Delta x/x) = \frac{1}{2}F\Delta x/(Ax) = \text{work} \div \text{volume}$. Multiply by the volume Ax for the energy (a common derivation question).
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- **Springs:** in **series** each spring carries the full force, so the total extension doubles and k halves. In **parallel** each carries half the force, so the extension halves and k doubles.
- **Stress-strain graph points:** limit of proportionality (end of the straight line) → elastic limit (beyond it the material does not return to its original length) → **yield point** (a sudden large plastic extension for little extra stress) → ultimate tensile stress (highest point) → breaking (fracture) point (end of the line).
- **Elastic deformation:** returns to its original shape/length when the force is removed. **Plastic:** permanently deformed. A spring loaded beyond the elastic limit has a permanent extension and a smaller stiffness.
- **Loading/unloading loop (rubber):** the area between the loading and unloading curves is the energy transferred to thermal energy (Jan 2024).

Stiff	Large Young modulus: a steep linear section (small strain for a given stress).
Strong	High breaking (ultimate) stress.
Ductile / malleable	Large plastic region: can be drawn into wires / hammered into shape. Ductile materials absorb energy as they deform plastically.
Brittle	Breaks with little or no plastic deformation (concrete, glass, cast iron).
Tough	Absorbs a lot of energy before breaking: large area under the stress-strain graph.
Hard	Resists indentation/scratching of its surface.
Concrete and steel	Concrete is strong under compression but weak in tension (about 40 MPa vs 10 MPa, Oct 2019). Steel rods take the tensile stress. Pre-stressed concrete stays under compression when a tensile force is applied.

YOUNG MODULUS EXPERIMENT (6-MARK METHOD IN JUNE 2024; PARTS IN MANY PAPERS)

1. Use a **long, thin** wire clamped at one end, with a marker on it and a metre rule alongside. A long, thin wire gives a large extension, so the percentage uncertainty is small. Add an initial load to keep the wire straight (taut).
2. Measure the original length x from the clamp to the marker.
3. Measure the diameter with a **micrometer** at several positions along the wire and at several orientations, then take the mean; $A = \pi d^2/4$.
4. Add masses; calculate the force $W = mg$; measure the extension (new reading – original), reading the scale at eye level to avoid parallax.
5. Calculate stress = F/A and strain = $\Delta x/x$, and plot stress against strain (or force against extension).
6. E = gradient of the straight section (or gradient $\times x/A$ for a force–extension graph).
Wear **safety goggles** in case the wire snaps.

A vernier scale (0.1 mm resolution) gives a smaller percentage uncertainty in the extension than a metre rule. A longer x keeps the same absolute uncertainty but a smaller percentage uncertainty.

10. Six-mark model answers

Each set contains six indicative points in order. Learn the chain; the context changes from paper to paper.

A. FALLING TO TERMINAL VELOCITY, THEN PARACHUTE. SKYDIVER: OCT 2020, JUN 2025; SIMILAR EVERY YEAR.

1. Initially the velocity is zero, so there is no air resistance and the acceleration is g (only weight acts).
2. As the velocity increases, air resistance (drag) increases.
3. So the resultant force ($W - D$) decreases and the acceleration decreases.
4. Eventually drag (+ upthrust) = weight, so the resultant force is zero and the skydiver reaches terminal velocity.
5. When the parachute opens, air resistance becomes greater than weight, so the resultant force is upwards and the skydiver decelerates.
6. As speed falls the drag decreases until it again equals the weight: a new, **lower** terminal velocity (drag equals weight at a lower speed because the area is larger).

Energy version (Jun 2025): E_{grav} decreases throughout; while accelerating E_k increases and work is done against air resistance; at terminal velocity E_k is constant and all the lost E_{grav} is dissipated.

B. BALLOON OR BUBBLE RISING (UPTHRUST). OCT 2022, OCT 2023, JUN 2026; FLOATING JAN 2019, JUN 2022.

1. When released, the upthrust is greater than the weight (and there is no drag yet),
2. so there is a resultant upward force and the balloon accelerates upwards (N2).
3. As the velocity increases, the air resistance (drag, downwards) increases; U and W are constant.
4. So the resultant force decreases and the acceleration decreases.
5. Eventually upthrust = weight + drag, so the resultant force is zero,
6. and the balloon rises at a constant (terminal) velocity.

Weather balloon variant: at altitude the air density falls, so upthrust decreases until it is less than the weight; the resultant force is then downwards and the balloon decelerates to rest.

Expanding bubble (Jun 2026): weight is constant; upthrust $\propto r^3$ increases faster than drag $\propto r$, so there is a resultant upward force and the bubble keeps accelerating. **Galileo thermometer (Jun 2022):** as temperature rises the liquid's density falls, so the upthrust on a bulb decreases until it is less than the bulb's constant weight, and the bulb sinks.

C. BUNGEE / ELASTIC ROPE: FORCES AND ACCELERATION. OCT 2024; PULL-BACK CAR OCT 2021.

1. Once the rope starts to stretch, initially the weight is greater than the tension, so the resultant force is downwards
2. and the acceleration is downwards.
3. As the rope extends, the tension increases (Hooke's law: tension $\propto$ extension).
4. So the resultant force decreases and the acceleration decreases.
5. At B the tension equals the weight, so the resultant force and the acceleration are zero (maximum speed).
6. Below B the tension is greater than the weight, so the resultant force is upwards and the jumper decelerates to rest.

Energy version (Jan 2021): stage 1: KE increases because gravity does work; stage 2: KE still increases but at a decreasing rate, because work is also done stretching the cord; stage 3: KE decreases because work is done on the cord faster than gravity does work on the jumper; at rest, total work done by gravity = work done stretching the cord.

D. NEWTON'S 2ND AND 3RD LAWS IN AN INTERACTION. BALL AND SKITTLE (JAN 2024), JET BALLOON (JUN 2023), ASTRONAUT SEAT (JUN 2021).

1. The ball (balloon) exerts a force on the skittle (the air).
2. The skittle (air) exerts a force on the ball (balloon).
3. By Newton's third law these forces are equal in size and opposite in direction.
4. So there is a resultant force on each object (for the balloon: forward force > resistive forces).
5. By Newton's second law each object accelerates in the direction of its resultant force.
6. So the skittle accelerates forwards and the ball decelerates. (Balloon: as speed increases the resistive force increases until it equals the forward force; then by N1 the acceleration is zero, so the balloon reaches maximum speed.)

E. SCALES IN A LIFT / TRAMPOLINE FORCES. JAN 2022, JAN 2023.

1. The reading on the scales is the normal contact (reaction) force of the scales on the student.
2. At constant speed (or at rest) the resultant force on the student is zero, so reaction = weight
3. and the reading equals the weight (600 N).
4. As the lift decelerates while moving up, the reaction is less than the weight,
5. so there is a resultant downward force on the student,
6. and the reading is less than 600 N.

Trampoline: weight acts down; the normal contact force acts up, is maximum at the bottom of the bounce and zero once contact is lost; resultant = difference between them; when the contact force > weight the acceleration is upward, and when it is less the acceleration is downward.

F. DESCRIBE THE MOTION FROM A DISPLACEMENT-TIME GRAPH. JAN 2026.

1. The gradient of a displacement-time graph represents velocity.
2. Up to 10 s the gradient is constant, so the velocity is constant (acceleration zero).
3. From 10 s to 20 s the gradient decreases, so the velocity decreases
4. and the box decelerates (negative acceleration).
5. At 20 s the gradient is zero, so the velocity is zero (the box is momentarily at rest).
6. After 20 s the gradient is negative, so the box moves back in the opposite direction.

G. OTHER CONTEXTS SET SO FAR (ONE-LINE CHAINS)

- **Submarine lifted out of water (Jan 2025):** resultant force zero, so tension = weight – upthrust; as it leaves the water the volume displaced and so the upthrust decrease; tension increases; out of the water, tension = weight.
- **Pendulum collision energy (Jun 2025):** as A falls, its GPE decreases and its KE increases and its momentum increases; in the collision momentum is conserved and energy transfers from A to E; E's KE and momentum fall to zero as its GPE rises; E's final GPE equals A's initial GPE (total energy conserved).
- **Power in stages (Oct 2025):** stage A: KE increases ($\propto v^2$), so the useful power increases; stage B: KE constant, GPE increases, steeper slope so power increases; stage C: GPE increases while KE decreases, so the person's useful power output is zero.
- **Projectile with air resistance (Oct 2019):** going up, weight + drag act down, so the deceleration is greater and the maximum height is lower; a horizontal drag force reduces the horizontal velocity, so the range is shorter.
- **Pull-back toy car (Oct 2021):** the wheels push back on the floor, so (N3) the floor pushes the car forward; the spring's compression reduces, so the force (Hooke) reduces, so the acceleration (N2) reduces to zero once the spring is relaxed.
- **Long thin wire (Jun 2019):** extension $\propto$ length and $\propto 1/\text{area}$, so a long thin wire gives a large extension, a lower percentage uncertainty, and allows smaller loads.
- **Comparing two stress-strain graphs (Jan 2020):** identical at low stress; the steeper graph is stiffer (greater E); larger maximum stress means stronger; larger area means more energy absorbed (tougher).

11. Short-answer phrase bank

Limit of proportionality	The point beyond which force is no longer proportional to extension (Hooke's law no longer applies; the graph stops being straight).
Elastic limit	The force/stress beyond which the material does not return to its original length when the force is removed.
Yield point	The stress at which a material undergoes a sudden, large plastic deformation.

Young modulus	Ratio of stress to strain (in the linear region), with symbols defined. It is not breaking stress, density or the elastic limit.
Upthrust	An upward force on an object in a fluid, equal to the weight of fluid displaced.
Why suvat cannot be used	The acceleration is not constant (the slope gradient varies or a resistive force changes), and the equations of motion require constant acceleration.
Unicycle at constant velocity	The resultant vertical force is zero, so there is no vertical acceleration. The forward friction balances the drag, so the horizontal resultant is zero and the speed is constant.
Diameter measurement	Micrometer screw gauge (or digital calliper) at several positions and orientations; take the mean.
Hammer and bell (Jan 2023)	The hammer head is not in free fall: the person exerts a force and does work on it, so more energy is transferred to KE than from gravity alone.
Faster car, larger force?	$P = Fv$: at the same power, a higher velocity (in higher gears) gives a smaller driving force.
Larger person, lower top speed	Greater cross-sectional area, so greater air resistance at a given speed.
Electromagnet drops a load (Oct 2023)	The change in elastic strain energy (trapezium area between the two weights) is greater than the change in GPE of the electromagnet (rectangle).

12. Multiple-choice traps seen in these papers

- N3 pairs: same type, different bodies, opposite directions, equal size, simultaneous. Weight and normal contact force on one object are never a pair.
- The N3 partner of “Earth pulls the cable car” is “the cable car pulls the Earth”.
- A car moving in a circle at constant speed: its momentum changes (direction) but its KE is constant.
- Area under a stress-strain graph = work per unit volume, not work.
- Use the radius, not the diameter, and convert mm $\rightarrow$ m before squaring or cubing.
- The gradient of s against t^2 is $g/2$; of v^2 against s it is $2g$.
- A falling object always has acceleration g (without air resistance), including at the top of its flight.
- Efficiency uses useful $\div$ total input; total input = useful + wasted.
- Area below the time axis of a v - t graph is negative displacement; area under a - t is the change in velocity.
- Doubling the diameter needs $4\times$ the tension for the same stress.
- Same material: breaking stress, density and E are independent of dimensions; the breaking force is not.
- Same stress: a longer wire breaks at a larger extension (same strain).
- Springs in series: extension adds; in parallel: it halves.
- Energy stored between forces F_1 and F_2 is the trapezium area $\frac{1}{2}(F_1 + F_2)\Delta x$.
- Larger temperature $\rightarrow$ lower viscosity $\rightarrow$ greater terminal velocity.

- Upthrust is constant for a fully submerged sinking object; only the drag changes.
- Coin and feather: same initial acceleration (g); the coin has a higher terminal velocity.
- Horizontal launch speed does not change the time to fall; vertical and horizontal motions are independent.
- Kinetic energy is not a vector; momentum and weight are.

13. Final checklist before the exam

Can you, without notes...	Can you, without notes...
Sort any quantity into vector/scalar and give its base units?	State conservation of momentum with the “no external force” condition?
Do a two-direction projectile and conclude against the target?	List all six properties of a Newton’s third law pair?
Draw a free-body diagram for a sphere at terminal velocity (W , U , D)?	Take moments with an angled force and find a hinge force?
Find η from $W - U = 6\pi\eta rv$ and state the four Stokes conditions?	Write the energy statement with work against resistive forces?
Find E from a force–extension graph using $E = \text{gradient} \times x/A$?	Describe the Young modulus and viscometer experiments step by step?
Explain terminal velocity in 6 linked steps (falling and rising)?	Define limit of proportionality, elastic limit, yield point and the material properties?

Data: $g = 9.81 \text{ N kg}^{-1} (\text{m s}^{-2})$; density of water $\approx 1000 \text{ kg m}^{-3}$ (seawater $\approx 1030 \text{ kg m}^{-3}$); $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$. Equations on the formulae sheet: suvat, $\Sigma F = ma$, $p = mv$, moment $= Fx$, $\Delta W = F\Delta s$, $E_k = \frac{1}{2}mv^2$, $\Delta E_{\text{grav}} = mg\Delta h$, $P = W/t$, efficiency, $\rho = m/V$, $F = 6\pi\eta rv$, $\Delta F = k\Delta x$, $E_{\text{el}} = \frac{1}{2}F\Delta x$, $\sigma = F/A$, $\varepsilon = \Delta x/x$, $E = \sigma/\varepsilon$.